

Robot Attention

How often should a robot check?

Name: _____

Date: _____

Follow the plan and keep checking

Walking, you follow a plan and watch for people or chairs. Robots need fresh information too: has a control changed? Is the path clear? Is the motor too hot? Frequent checks can help avoid collisions and damage.

Today we check the RV control. Obstacles and heat would need other sensors.

Lesson 04: Check, act, and wait

The `loop()` repeats: **read** A0 (0–1023), **map** to PWM (0–255), **apply** on D3, **report** the numbers, then **wait**.

Code pieces, in order:

```
analogRead(ANALOG_INPUT_PIN)
map(adcReading, ADC_MIN, ADC_MAX,
    PWM_MIN, PWM_MAX)
analogWrite(MOTOR_PWM_PIN, pwmCommand)
Serial.print(adcReading);
Serial.println(pwmCommand);
delay(CONTROL_UPDATE_DELAY_MS);
```

The delay is **1000 ms: one second**. It is **blocking**: no new RV reads during the wait. The motor keeps its old PWM power setting.

Move the RV just after a reading: the command can lag by **about one second**. Just before the next reading, the wait is shorter. An old “go” command could cause a collision.

Lesson 05: A timer reminds the robot to check

Every **100 ms** (one tenth of a second), a timer **interrupt** pauses regular work for a brief checking routine: the **ISR**. Then regular work continues.

Inside `ISR(TIMER1_COMPA_vect):`
selected code pieces, in order

```
analogRead(ANALOG_INPUT_PIN)
map(adcReading, ADC_MIN, ADC_MAX,
    PWM_MIN, PWM_MAX)
analogWrite(MOTOR_PWM_PIN, pwmCommand)
```

It reads, maps, and applies the PWM power setting **10 times a second**, even with the RV still. PWM's fast on/off pulses have a separate rhythm.

`loop()` still waits with `delay(REPORT_DELAY_MS);` (1000 ms). Both print about once a second. ISR checks continue, updating power **before the next computer message**.

Compare the checks: one second

• = read the RV and update the power setting. Each box spans 100 ms.

Time (ms)	0	100	200	300	400	500	600	700	800	900
Lesson 04	•									
Lesson 05	•	•	•	•	•	•	•	•	•	•

1000 ms starts the next second. This sketch shows the checking rhythm; real timing is approximate.

Plan the Experiment

Write your hypothesis before testing the two programs.

Name: _____ Date: _____

Our investigation question

Will the motor take longer to respond to an RV change when checks are 1000 ms apart than when they are 100 ms apart?

We will listen to and watch the motor to find out.

My hypothesis and reason

A **hypothesis** is a prediction we can test. Before running either program, write what you think will happen and why, in your own words.

You can begin: "I think the motor will respond ... because ..."

Experimental steps

1. **Prepare.** Use the same B1, RV, and motor circuit. Remove USB and B1 power before wiring. Have your teacher check it. Secure the bare motor with no wheel or propeller; keep its shaft clear.
2. **Run lesson 04.** With B1 off, follow its README to build, upload, and open the monitor at 9600 baud.
3. **Set the RV.** With B1 off, choose a low position near zero power and a high position. Return to low; turn B1 on after teacher approval.
4. **Test and record on page 3.** Just after a computer message, move low to high and hold. Notice how soon the motor responds. Repeat high to low, then low to high, each just after a message.
5. **Run lesson 05.** Turn B1 off; leave the wires unchanged. Follow its README to build, upload, and reopen the 9600-baud monitor.
6. **Repeat.** With B1 off, return to low. After teacher approval, turn B1 on. Repeat the same three moves just after messages; record on page 3.
7. **Finish safely.** Turn B1 off and disconnect USB after closing the monitor. Stop and tell your teacher if anything gets hot or acts unexpectedly.
8. **Conclude.** Compare the trial records. Use what you noticed to answer page 3. "Hard to tell" is a useful observation too.

Record Results and Conclude

Name: _____ Date: _____

My observations

Describe what you heard or saw: “noticeable wait,” “quick change,” or “hard to tell.”

Trial and RV move	Lesson 04 1000 ms checks	Lesson 05 100 ms checks
1: Low to high		
2: High to low		
3: Low to high		

My conclusion

Did the 100 ms version (lesson 05) respond sooner to your RV moves?

Circle: **Yes** / **No, about the same** / **No, slower** / **Mixed results** / **Hard to tell**

What evidence from your trial records supports your answer?

Did your results support your hypothesis? Explain, or say why you are unsure.

Explain the checking

What power setting does lesson 04 keep sending during its wait?

Checking every 100 ms: $1000 \div 100 = \underline{\hspace{2cm}}$ checks per second.

Think about a robot problem

Choose a problem: an obstacle could cause a collision and damage, a control instruction changes, or a motor becomes too hot. What information should the robot check, and how could frequent checks help?
